

AcerbE™ Security Model Overview

Threat Model, Attack Surface, Mitigations, and Cryptographic Primitives

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1. SECURITY POSTURE

AcerbE™ is a zero-trust, local-only integrity engine. No external network connections are established. No data leaves the operator's system. All cryptographic operations are deterministic and verifiable.

2. THREAT MODEL

Threat	Vector	AcerbE™ Response
Unauthorized distribution	Buyer shares product file	Fingerprint identifies buyer. Registry provides provenance.
File scrubbing	Attacker strips metadata	LSB embedding survives metadata stripping. Multiple layers redundant.
File modification	Attacker alters file bytes	HMAC-SHA256 tamper-guard fails. Output is zero-byte and unusable.
Registry tampering	Attacker modifies registry	Each record sealed with HMAC. Modification is detectable.
Key extraction	Attacker extracts buyer key	Keys are PBKDF2-derived with 200,000 iterations. Not stored in file.
Format conversion	Buyer converts to different format	LSB embedding survives PNG re-save. PDF metadata survives re-export.
Brute force identification	Third party scans for fingerprint	Encrypted payload requires buyer-specific derived key to decode.
Replay attack	Attacker reuses fingerprint block	Each block is unique per UUID + transaction. Replay is detectable.

3. CRYPTOGRAPHIC PRIMITIVES

Primitive	Algorithm	Standard	Purpose
Symmetric encryption	AES-256-CBC	FIPS 197	Payload encryption
Key derivation	PBKDF2-HMAC-SHA256	NIST SP 800-132	Buyer-unique key generation
Integrity verification	HMAC-SHA256	FIPS 198-1	Tamper detection
File hashing	SHA-256	FIPS 180-4	Original file provenance

Padding	PKCS7	RFC 5652	Block cipher padding
IV generation	os.urandom(16)	CSPRNG	Unique IV per encryption

4. ATTACK SURFACE

Surface	Exposure	Mitigation
Registry database	Local filesystem only	Operator controls filesystem permissions
Master key	Environment variable	Never hardcoded. Operator responsibility.
Stamped output files	Delivered to buyer	Encrypted. HMAC-sealed. Tamper-destructive.
Local server (port 7749)	Localhost only (127.0.0.1)	Not accessible from network.
Temp upload directory	Local filesystem	Cleared after processing.
Python runtime	Local execution	Standard OS process isolation.

5. OPERATOR SECURITY OBLIGATIONS

- Set ACERBE_MASTER_KEY to a cryptographically random 256-bit value via environment variable. Never use the default.
- Set ACERBE_REGISTRY_KEY to a separate cryptographically random value.
- Restrict filesystem permissions on acerbe_registry.db to the operating user only (chmod 600 on Unix).
- Back up acerbe_registry.db to a secure, access-controlled location.
- Log and retain buyer_key_hex values in a separate secure store.
- Do not expose port 7749 to any network interface. AcerbE™ binds to 127.0.0.1 only.
- Keep Python and all dependencies updated to current stable releases.

6. LIMITATIONS

- AcerbE™ does not prevent a determined attacker with unlimited compute from brute-forcing a derived key if the master key is weak or exposed.
- LSB steganography in images does not survive lossy JPEG re-compression at high compression ratios.
- AcerbE™ outputs are integrity artifacts, not court-certified forensic evidence. Independent expert verification is required for legal proceedings.
- AcerbE™ does not prevent a buyer from simply re-creating a product manually from memory or notes.